DRAFT Case Brief for EVNew Case v5.1

Author's note:

This case is inspired by the rotation of valuation methods as market prices evolve over a business cycle. Stocks at/near their cyclical troughs tend to have a value-oriented investor base, with the investor base converting to a growth bias as the stock price moves closer to a cyclical peak.

You are working in the active equities group at a large pension fund manager. Your group covers a number of large capitalization stocks and you have been assigned coverage of Teldar Paper (TA), Anacott Steel (AS) and Bluestar Airlines (BA). From your preliminary work on these companies, you have determined that investors apply different valuation methodologies to forecast future prices and you are tasked with making trading decisions for these stocks over the next year.

The investor base has two constituents, institutional investors and retail investors. Institutional investors follow a value style for TP and AS and utilize discounted cash flow models for valuing these two stocks, in particular the 2-Stage Dividend Discount Model (DDM). Retail investors have a growth bias and favour multiples for valuation, particularly the Price-to-Earnings (P/E) ratio, for valuing all three stocks. Institutional investors use a modified P/E ratio to value BA, for which they raise/lower their P/E ratio based on BA's growth/decline in Earnings-Per-Share (EPS). An analysis of historical prices indicates that stock prices are approximately equal to the weighted average valuation of the institutional and retail investors, where the weights represent the proportion of each stock owned by each type of investor.

Your firm's research department has prepared the following summary of EPS for last year:

	Q1	Q2	Q3	Q4	Full Year
Teldar Paper EPS (Last Year Actual)	0.40	0.33	0.33	0.37	1.43
Anacott Steel EPS (Last Year Actual)	0.35	0.45	0.50	0.25	1.55
Bluestar Airlines EPS (Last Year Actual)	0.15	0.50	0.60	0.25	1.50

In addition, your research department has estimated that the costs of equity for TP, AS, and BA are 5%, 7.5%, and 10%, respectively. Retail investors value TP, AS, and BA with P/E ratios of 12, 16, and 20, respectively. Institutional investors apply a P/E ratio of 20 to BA, modified by BA's EPS growth (e.g. if BA's EPS growth is +5%, the Institutional P/E ratio would be 20 * (1 + 5%) = 21). Both types of investors value the companies based on trailing 12-month, or Last-Twelve-Months (LTM), EPS. TP pays out 80% of its EPS as a dividend and AS has a dividend payout ratio of 50%. Bluestar Airlines does not pay a dividend.

Institutional investors utilize the following 2- Stage DDM:

$$DDM = PV_{Years\ 1\ to\ 5} + PV_{Terminal\ Phas}$$

Stage 1: Present Value of dividends expected over the next 5-years:

$$PV_{Years\ 1\ to\ 5} = \frac{Year\ 1\ Dividend}{Cost\ of\ Equity - Divident\ Growth\ Rate} \times \left[1 - \left(\frac{1 + Dividend\ Growth\ Rate}{1 + Cost\ of\ Equity}\right)^5\right]$$

Stage 2: Present value of dividends expected from year 6 to infinity:

$$PV_{Terminal\ Phase} = \frac{\left[\frac{Year\ 6\ Dividend}{Cost\ of\ Equity - Terminal\ Growth\ Rate}\right]}{(1 + Cost\ of\ Equity)^5}$$

The Dividend Growth Rate is calculated as the change in LTM dividends from the prior year. At the end of the upcoming year (i.e. the end of the case), the Dividend Growth Rate will be the percentage change between the dividends generated from reported earnings during the case and the dividend paid in the prior year based on the actual EPS from last year. The initial expectation is that the dividend growth rate will be zero (EPS for the upcoming year is expected to the same as the prior year). The Terminal Growth Rate for dividends is 2% for all companies.

The companies will publicly release quarterly earnings at ticks 60, 120, 180, and 240. The case will end on tick 300, just before the next scheduled quarterly earnings release. Shortly before each earnings release, you will receive private earnings estimates from an analyst. The analyst will provide estimates for the remaining quarterly EPS yet to be released. Due to the greater visibility for near-term earnings over more distant earnings, the analyst estimates will fall within a range around the actual quarterly EPS depending on the amount of time before each earnings release. Analyst ranges are uniformly distributed +/- 0.02 (TP) / 0.04 (AS) / 0.06 (BA) per quarter of time until the earnings are released. For example, at the start of the case the Q4 estimates have a range of +/- 0.08 around the actual Q4 EPS for TP, and Q3 estimates have a range of +/- 0.12 around the actual Q3 EPS for AS.

Shortly after earnings are released each quarter, your analyst will also provide a private estimate of the proportion of shares owned by institutional investors, based on information your analyst can glean from various quarterly reports filed by institutional investors. As the year progresses, the estimate of institutional investor improves. Initially, starting with Q1 earnings, your analyst can only be sure that the true proportion of institutional ownership is within +/- 20% (TP) / 25% (AS) / 30% (BA) of their estimate (uniformly distributed). After each quarter, your analyst's accuracy improves and the range around the true proportion shrinks by 5%. For example, in Q2 the ownership proportion is within +/- 15% of your

analyst's estimate for TP and in Q3 the range falls within +/- 10% for TP. The institutional ownership proportion is uniformly distributed with a mean of 50%.

At the end of case, all positions are closed out at the final settlement value:

For TP and AS

$$\begin{aligned} &Final\ Settlement\ Value \\ &\qquad = Institutional\ Ownership \times DDM \\ &\qquad + (1 - Institutional\ Ownership) \times \frac{P}{E}\ Valuation \end{aligned}$$

For BA

$$\begin{aligned} &Final\ Settlement\ Value \\ &\qquad = Institutional\ Ownership \times Institutional \frac{P}{E} Valuation \\ &\qquad + (1 - Institutional\ Ownership) \times Retail\ \frac{P}{E}\ Valuation \end{aligned}$$

There is a $0.01 trading commission per share, and your trading desk has hard position limits of 100,000 gross shares and 50,000 net shares. Maximum order size for each stock is 10,000.